

1. Generative AI-Based Automated Analysis and Risk Scoring of Fraudulent APKs

Problem Description

Fraudsters increasingly distribute malicious mobile applications (APKs) through platforms such as WhatsApp, SMS, email, and phishing links to steal customer credentials, access sensitive information, and perform unauthorized financial transactions. Manual analysis of such APKs is complex, time-consuming, and dependent on skilled cybersecurity experts.

The proposed solution aims to develop a ***Generative AI-powered malware analysis system*** capable of automatically analyzing suspicious APK files and identifying malicious behavior. The system should leverage GenAI for reverse engineering, malware pattern recognition, automated code interpretation, and intelligent threat summarization, along with static and dynamic analysis techniques to examine application permissions, APIs, embedded code, runtime activities, and network communications.

Using AI-driven insights, the solution should detect malware patterns, classify threat severity, generate risk scores, and produce detailed investigation reports with actionable recommendations. The objective is to enable faster identification of fraudulent applications and support proactive cybersecurity and fraud prevention measures for banks.

2. AI/ML-Based Classification of Suspicious Mule Accounts

Problem Description

Banks are facing a growing number of cyber-enabled financial frauds involving mule accounts used to receive, transfer, and conceal fraudulent funds across multiple banking channels. Traditional rule-based monitoring systems are often unable to identify evolving fraud patterns and suspicious account behavior in real time.

The proposed solution aims to develop an ***AI/ML-powered classification system*** capable of ***identifying suspicious and mule accounts*** by analyzing features from financial transaction data ***provided in this portal***.

Using the data provided, the solution should build AI/ML models that learn behavioural and transactional patterns to identify ***suspicious and mule accounts***. The system should leverage machine learning techniques for anomaly detection, predictive risk scoring, and intelligent alert generation to help banks proactively detect and prevent the misuse of mule accounts in fraudulent fund movements.

The objective is to perform feature engineering on this data to identify the most relevant features for fraud and mule account detection, and build a classification model to accurately distinguish suspicious accounts from legitimate ones

Note: In the dataset, feature 3924 is the target variable. The following are some commonly used features utilized by bank for fraud detection.

F115

F321

F527

F531

F670

F1692

F2082

F2122

F2582

F2678

F2737

F2956

F3043

F3836

F3887

F3889

F3891

F3894